

Forensic Data Analysis

CYFI 445

Instructor

Dr. Frank (Weifeng) Xu

E-mail: wxu@ubalt.edu

Office: LAP 526 **Phone:** 410-837-5302

Office hours: By appointment

Class Information

Credits: 3 credits

Location: BC 015 (in person); Zoom (online)

Zoom: [Join the online class](#)

<https://ubalt.zoom.us/j/5547149563>

Course Description

This course introduces concepts and techniques in data analytics and analysis for discovering forensic evidence. Students apply basic statistical, machine-learning, and artificial-intelligence tools to describe, visualize, and analyze forensic data collected from computing devices. The course focuses on detecting anomalies in collected forensic log files.

Lab fee: \$65.

Prerequisites: CYFI 305, CYFI 315, CYFI 345, CYFI 360, and CYFI 415.

Meeting Format

The course meets every Monday and is offered in two formats:

- **Hybrid (HYB):** In-person attendance on campus is required on September 14, September 28, October 12, and October 26; all remaining lectures will be held remotely via Zoom.
- **Zoom (ZM):** Remote attendance via Zoom is required for all class sessions.

Student Learning Outcomes

- Describe data analytics and its role in forensic investigations.
- Apply basic statistical concepts to interpret the characteristics of forensic data.
- Apply basic tools to carry out data-visualization tasks.
- Describe basic data-mining algorithms and select appropriate algorithms for predictive modeling.
- Apply basic data-mining, machine-learning, and deep-learning algorithms and tools to detect anomalies in forensic log files.
- Evaluate forensic analysis results.

Textbook and Learning Resources

- **No required textbook:** <https://www.learnpytorch.io/>
- <https://github.com/frankwxu/agent-ic-AI4-forensics>
- <https://github.com/frankwxu/AI4DigitalForensics/tree/main/lectures>
- <https://github.com/dair-ai/ML-YouTube-Courses>

Course Topics (Tentative)

- Python Basics
- LLM Foundations
- Environment Setup
- LLM API and Model Basics
- AI Agents
- Reflection Pattern
- Tool-Use Pattern

- ReAct Pattern
- Planning Pattern
- Multi-Agent Pattern

Grading Criteria

Lab/Projects/Homework/Quizzes: 200 points

Midterm: 100 points

Final: 100 points

Total: 400 points

Grade thresholds: A: 95%; A-: 90%; B+: 87%; B: 83%; B-: 80%; C+: 77%; C: 73%; C-: 70%; F: below 70%.

Grade policy

- Late homework/quizzes/labs will **not** be graded for **any** reason.
- Students who miss two class sessions will fail the course.
- No make-up quizzes/exams will be allowed without prior arrangements being made.
- When emailing questions, capture screenshots and explain how you verified every command used, or the email will be ignored.
- To appeal a grade, email the instructor within two weeks of receiving the grade. Overdue appeals will not be considered.
- If the total points do not equal 400, grades will be calculated as a percentage.

Appeal a Grade

When appealing a grade, it is important to provide a clear and well-justified reason for why the grade should be re-evaluated. Use the following template and include relevant supporting reasons.

[Your Name]
 [Your Student ID]
 [Course Name and Number]
 [Date]

1. **Detailed Explanation of the Issue:** Clearly and concisely describe the specific aspect(s) of your work that you believe were not accurately assessed or graded. Be as specific as possible and refer to relevant sections of your assignment or exam.
2. **Providing Supporting Evidence:** Attach or reference relevant materials, such as rubrics, assignment guidelines, or notes from class discussions, that demonstrate your understanding of the subject matter and your adherence to the assignment requirements.
3. **Comparative Analysis:** Compare your work to the grading criteria and any exemplary solutions or model answers that were provided. Highlight areas where you believe your work meets or exceeds the expectations outlined in the course.
4. **Addressing Feedback:** If you received feedback from the initial grading, explain how you have incorporated that feedback into your work and why you believe it should lead to a higher grade.
5. **Clarity in Communication:** Ensure your appeal is well-organized, concise, and respectful in tone. Clearly articulate your points and avoid unnecessary emotional language.
6. **Meeting Deadlines:** Check the institution's policies regarding grade appeals and deadlines. Make sure your appeal is submitted within the specified timeframe.

Other Important Information

Academic Calendar and Important Dates

<https://www.ubalt.edu/academics/classes-and-registration/academic-calendar.cfm>

Recording Statement

Faculty may be required to record classes for the purposes of accommodating a disability or may opt to do so for students who cannot attend or who wish to review the full class content. All recordings are for the sole use of class instruction and study and may not be reproduced by students for any other purpose; to do so is a conduct violation. Faculty cannot reproduce students' voices or images from the class for any other purpose without additional student consent. All such recordings are protected by a UB login process based on where they are posted. Students may mute their microphone or turn off their camera if they do not consent to be recorded, but this may mean they need to find additional ways to participate in class. Faculty are to notify students when a class is recorded.

Students can use a virtual background on Zoom to protect their privacy while remaining clearly in attendance and engaged in class. Students should keep in mind that faculty have to be able to determine if a student is truly participating in a class to comply with University and federal attendance policies. A student could be deemed absent if logged in to a synchronous class but not responding. Visual and/or audio presence may be required for examinations or other types of assessment.